Imperial College London
Department of Earth Science and Engineering
MSc Environmental Data Science and Machine Learning

High-Resolution Spatial Downscaling of Wildfire Data

Author:
Tsuyu Huang
tsuyu.huang25@imperial.ac.uk

Supervisors:
Jorge Veiras (jorge@artiocarbon.com)
Prof. Anna Korre (a.korre@imperial.ac.uk)

Code Repository:
https://github.com/ese-ada-lovelace-2025/irp-th25

Company:
Artio
www.artiocarbon.com

May 28, 2026

Abstract

Wildfires are destructive and a rapidly evolving hazard. With climate change, they become increasingly severe and frequent. This increases the need for a better wildfire risk prediction and monitoring using satellite data. This project aims to achieve a high-resolution spatial downscaling of global wildfire data.

Contents

1 Introduction **3**

2 Materials **3**

3 Methods **3**

4 Results **3**

5 Discussions **3**

6 Conclusions **3**

1 Introduction

Introduction text goes here. Citations should be included as needed, for example (Badhan et al., 2024).

2 Materials

3 Methods

4 Results

5 Discussions

6 Conclusions

References

Badhan, M., Shamsaei, K., Ebrahimian, H., Bebis, G., Lareau, N. P., & Rowell, E. (2024). Deep Learning Approach to Improve Spatial Resolution of GOES-17 Wildfire Boundaries Using VIIRS Satellite Data. *Remote Sensing*, *16*(4), 715. https://doi.org/10.3390/rs16040715